

Lab Assignment 5: Exploring the Properties of a Continuous Probability Distribution (Normal Distribution)

Problem Statement:

You are given a Normal distribution with a mean (μ) = 50 and standard deviation (σ) = 10. Using Python (NumPy, SciPy, and Matplotlib), perform the following tasks:

Sub-Questions:

(a) Visualization and Theoretical Properties

1. Plot the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of the distribution.
2. Identify and print the mean, variance, and standard deviation using theoretical formulas.

(b) Sampling and Empirical Statistics

1. Generate 1000 random samples from the given distribution.
2. Plot a histogram of the samples.

(c) Probability Computation

1. Compute the probability that a randomly selected value is between 40 and 60.
2. Compute the probability that a value is greater than 65.